

Daniel Yan

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EDUCATION

University of Toronto

BASc in Mechanical Engineering + PEY Co-op

- GPA: 3.49 — Dean's Honour List

Toronto, ON

Sep 2024 – Expected May 2029

SKILLS

Technical Skills: SolidWorks, Ansys, CATIA, Abaqus, CATIA Composites Design, 3DEXPERIENCE, Python

Tools: MS Office, Slack, Git, Gantt Chart

Certification: CSWA, G2 Driver's License, George Brown Machining (lathe, mill), Mandarin (Fluent), English (Native)

EXPERIENCE

Structural & Fabrication Lead

Toronto, ON

Blue Sky Solar Racing – UofT Design Team

Apr 2025 – Current

- Optimized GEN12's bottom aerobody ply schedule via CATIA Composites Design with a skeletal core design; achieved a **41% mass reduction** while using Ansys ACP to validate structural limits under Bernoulli aerodynamic pressures
- Designed a roll cage in CATIA, leveraging topology optimization over **17 iterations** to meet race regulations, achieving **35% mass reduction** while maintaining an average minimum **safety factor of 1.2** under **10** regulatory load cases
- Spearheaded the adoption of **CATIA Composites Design** to optimize ply layup precision and streamline CAD-to-Ansys FEA workflows, reducing scrap material weight from excessive ply overlaps; facilitated knowledge transfer to **20+** members via a self-authored **50+ page** technical manual to standardize tool proficiency
- Led end-to-end development of a driver ergonomics test platform to optimize cockpit layout, ingress/egress biomechanics, and safety system integration; translated multi-driver measurements into chassis design inputs, reducing egress time by **5 seconds** under race conditions
- Designed a ventilated paint booth for component finishing, ensuring all specifications were met for Environmental, Health, and Safety (**EHS**) **compliance**; managing a Bill of Materials achieving a **53%** cost saving over retail models

Composites Intern

Toronto, ON

Multifunctional Composites Manufacturing Lab – University of Toronto

May 2026 – Current

- Synthesized a composite material with enhanced mechanical properties by hot-pressing **1,000+ alternating layers** of pure polypropylene and 0.5% graphene nanoplatelets, analyzing behaviors for semiconductor applications
- Engineered carbon fiber laminates infused with varying weight percentages of Halloysite Nanotubes (HNT) to maximize matrix hardness, reducing structural material requirements to optimize weight reduction in exoskeleton frameworks

Server & Busser

Toronto, ON

Gyubee Japanese Grill – Dundas

May 2026 – Current

- Adapted rapidly to a high-volume, fast-paced restaurant environment, efficiently managing front-of-house customer communication, taking accurate orders, and maintaining attentive table service under peak dining pressure
- Executed fast turnover cleaning, table resets, and utensil preparation, while managing end-of-day restaurant setup and inventory staging to ensure seamless, high efficiency operations for the following shift

Senior Business Member – Photography

Toronto, ON

University of Toronto Formula Racing – UofT Design Team

Dec 2025 – Current

- Independently coordinated and led photoshoots for **10+** corporate sponsors, directing subjects, managing shoot logistics, and maintaining professional relationships to deliver sponsor-ready promotional content on deadline
- Facilitated on-site photography sessions featuring the race car and sponsor products, proactively engaging stakeholders and team members to produce polished, brand aligned visual content for each partner

PROJECTS

Badminton Shuttlecock Launcher

May 2025 – Current

SolidWorks, ESP32, Electrical Design, Systems Integration, C++

- Designed and engineered a **fully-automatic 2-DOF** launching system in SolidWorks with pan/tilt capabilities to cover the entire badminton court, featuring adjustable trajectory height up to **4 meters** and maximum shuttlecock **speed of 180 km/h** for realistic **single-player** training scenarios
- Integrated electromechanical systems including high-speed brushless motors with ESCs, servo motors, ESP32, and a LiPo battery; implemented PWM control and fault-tolerant wiring to ensure consistent performance
- Developed **custom C++ control software** with HTML-based Bluetooth interface for iPhone connectivity, with fully customizable training sequences for various drill types